

A quick comment about Maxwell-Boltzmann Distrib

• Start with $E = \frac{1}{2} M(v_x^2 + v_y^2 + v_z^2)$

• Choose a direction x plus component in that direction into $P(E) \propto e^{-E/k_B T}$

To get $g(v_x) \propto e^{-Mv_x^2/2k_B T}$ where $g(v_x) dv_x$ is the fraction of molecules with v_x in x direction between v_x & $v_x + dv_x$.

• Leads to a normal distrib which when normalised gives

$$g(v_x) = \sqrt{\frac{m}{2\pi k_B T}} e^{-Mv_x^2/2k_B T}$$

• $\langle v_x \rangle = 0$, $\langle |v_x| \rangle = 2 \int_0^\infty v_x g(v_x) dv_x = \sqrt{\frac{2k_B T}{\pi m}}$

$$\langle v_x^2 \rangle = \int_{-\infty}^\infty v_x^2 g(v_x) dv_x = k_B T / M$$

• Fraction between v_x & $v_x + dv_x$, v_y & $v_y + dv_y$, v_z & $v_z + dv_z$

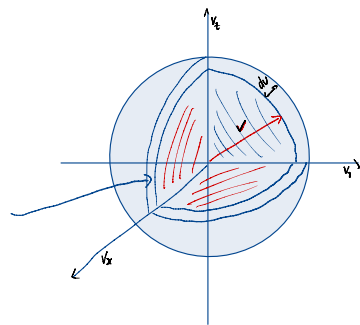
is $g(v_x) dv_x g(v_y) dv_y g(v_z) dv_z \propto e^{-Mv^2/2k_B T} dv_x dv_y dv_z$

Speed distribution

Want to work out fraction $f(v)$ between $v = |v|$ & $v + dv$.

Volume of the shell of velocity space of radius v & thickness dv is

$$4\pi v^2 dv$$



• therefore $f(v) dv \propto v^2 dv e^{-Mv^2/2k_B T}$

$$f(v) dv = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2k_B T} \right)^{3/2} v^2 e^{-Mv^2/2k_B T}$$

$$\langle KE \rangle = \frac{1}{2} M \langle v^2 \rangle = \frac{3}{2} k_B T$$

Energy and the first law

Equilibrium states $\rightarrow$ particular set of stable macroscopic observables
sudden addition of heat might change this $\sim$ initially at least.

"Functions of state" $\rightarrow$ a physical quantity that has well defined value for each equilibrium state
(or "variables")

Examples: Volume, pressure, Temperature, Internal Energy

Not examples: Position of a particle, total work done, total heat added.

Why not work and heat? Warming hands analogy $\rightarrow$ Temperature (function of state)
• independent of how you warmed your hands.

* Precise definition

• Let the state of the system be described by parameters $x = (x_1, x_2, \dots)$ and let $f(x)$ be some function of state. (e.g. $f(x) = x_1$)

• Then if x changes from x_i to x_f then

$$\Delta f = \int_{x_i}^{x_f} df = f(x_f) - f(x_i) \rightarrow \text{only depends on end points } x_i \text{ \& } x_f.$$

• This is called an exact differential. (see B&B. C7)

$F_1(x, y) dx + F_2(x, y) dy$ is an exact differential if it can be written as $df = \left(\frac{\partial f}{\partial x}\right) dx + \left(\frac{\partial f}{\partial y}\right) dy$ of a differentiable single valued function $f(x, y)$.

$$\Rightarrow F_1 = \left(\frac{\partial f}{\partial x}\right), F_2 = \left(\frac{\partial f}{\partial y}\right) \Leftrightarrow \vec{F} = \nabla f.$$

Properties: $\int_1^2 F_1 dx + F_2 dy = \int \vec{F} \cdot d\vec{r} = \int df = f(2) - f(1).$

$$\oint F_1(x, y) dx + F_2(x, y) dy = \oint \vec{F} \cdot d\vec{r} = \oint df = 0 \Rightarrow \nabla \times \vec{F} = 0$$

so $\left(\frac{\partial F_2}{\partial x}\right) = \left(\frac{\partial F_1}{\partial y}\right)$ or $\left(\frac{\partial^2 f}{\partial x \partial y}\right) = \left(\frac{\partial^2 f}{\partial y \partial x}\right)$

Stokes Theorem.

Example 11.1

Let a system be described by two parameters, x and y . Let $f = xy$ so that

$$df = d(xy) = y dx + x dy. \quad (11.2)$$

Then if (x, y) changes from $(0, 0)$ to $(1, 1)$, the change in f is given by

$$\Delta f = \int_{(0,0)}^{(1,1)} df = [xy]_{(0,0)}^{(1,1)} = (1 \times 1) - (0 \times 0) = 1. \quad (11.3)$$

This answer is independent of the exact path taken (it could be any of those shown in Fig. 11.1) because df is an exact differential.

Now consider² $dg = y dx$. The change in g when (x, y) changes from $(0, 0)$ to $(1, 1)$ along the path shown in Fig. 11.1(a) is given by

$$\Delta g = \int_{(0,0)}^{(1,1)} y dx = \int_0^1 x dx = \frac{1}{2}. \quad (11.4)$$

However when the integral is not carried out along the line $y = x$, but along the path shown in Fig. 11.1(b), it is given by

$$\Delta g = \int_{(0,0)}^{(1,0)} y dx + \int_{(1,0)}^{(1,1)} y dx = 0. \quad (11.5)$$

If the integral is taken along the path shown in Fig. 11.1(c), yet another result would be obtained, but we are not going to attempt to calculate that!

Hence we find that the value of Δg depends on the path taken, and this is because dg is an inexact differential.³

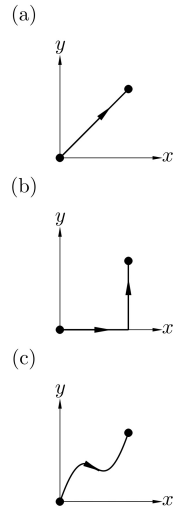


Fig. 11.1 Three possible paths between the points $(x, y) = (0, 0)$ and $(x, y) = (1, 1)$.

²We put a line through quantities such as the d in dg to signify that it is an inexact differential.

Notation point $\longrightarrow$

Functions of state : Extensive : Intensive
 Volume : pressure
 Energy : Temperature
 Mass : energy density

Equation of state : $f(p, V, T) = 0$ e.g. $pV - Nk_B T = 0$

The first law of thermodynamics

- Energy is conserved & both heat and work are forms of energy.
- A system has internal energy U ... all the internal degrees of freedom that the system possesses
- We can change the internal energy by heating it or by doing work on it.

$$\Delta U = \Delta Q + \Delta W$$

$\left\{ \begin{array}{l} \text{heat supplied to} \\ \text{work done on the system} \end{array} \right\}$
 Not functions of state as they are to do with how energy got into the system.

* A thermally isolated system $\Rightarrow \Delta Q = 0 \Rightarrow \Delta U = \Delta W$

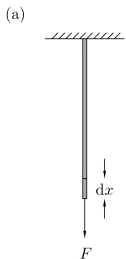
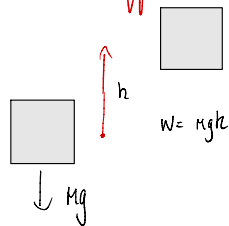
* For a small differential change in the system we write

$$dU = dQ + dW$$

is exact differentials

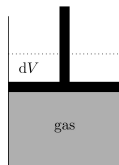
* Definition of work?

$$W = F \cdot d$$

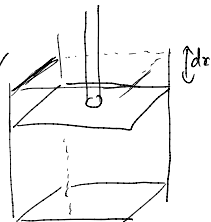


$$dW = F dx$$

(b)



$$\begin{aligned}
 dW &= -p dV \\
 &= \frac{F}{A} \cdot A \cdot dx \\
 &= F \cdot dx
 \end{aligned}$$



* Minus sign as work done > 0 when $dV < 0$

* Heat capacity revisited

$$\Delta U = \Delta Q + \Delta W$$

$$\Delta Q = \Delta U - \Delta W$$

$$C = \lim_{\Delta T \rightarrow 0} \frac{\Delta Q}{\Delta T} = \lim_{\Delta T \rightarrow 0} \frac{\Delta U - \Delta W}{\Delta T}$$

$$= \frac{dU}{dT} - \frac{dW}{dT}$$

$$= \frac{dU}{dT} - \frac{dW}{dV} \left(\frac{dV}{dT} \right)$$

$$= \frac{dU}{dT} + P \left(\frac{dV}{dT} \right)$$

$$W = -\int p dV$$

$$\frac{dW}{dV} = -P$$

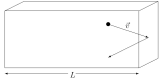
Constant Volume

$$C_V = \left(\frac{dQ}{dT} \right)_V = \left(\frac{dU}{dT} \right)_V + P \left(\frac{dV}{dT} \right)_V = 0 = \left(\frac{dU}{dT} \right)_V$$

Constant pressure

$$C_P = \left(\frac{dQ}{dT} \right)_P = \left(\frac{dU}{dT} \right)_P + P \left(\frac{dV}{dT} \right)_P$$

Quick reminder of ideal gas



- + Consider the pressure exerted by a single molecule
- + Exerts a force only when it bounces off the wall

• Elastic collision so $F_x = -\frac{\Delta p}{\Delta t} = -m \frac{\Delta v_x}{\Delta t} = -m \frac{[-v_x - v_x]}{\Delta t} = + \frac{2mv_x}{\Delta t}$

How long do we have to wait for it to happen again?

→ Δt is the time it takes to bounce off left wall again & back again, so $\Delta t = 2L/v_x$

→ plugging in we get $F_x = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}$

→ The pressure $P_{(x)} = \frac{F}{A} = \frac{mv_x^2}{AL} = \frac{mv_x^2}{V} \leftarrow \text{Volume}$

• Assume gas is isotropic → same in all directions →

$\langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = \frac{1}{3} (\langle v_x^2 \rangle + \langle v_x^2 \rangle + \langle v_x^2 \rangle) = \frac{1}{3} (3 \cdot \langle v_x^2 \rangle) = \langle v_x^2 \rangle$

which rearranges to $\langle v^2 \rangle = \frac{3pV}{Nm} = \frac{3p}{\rho} = \frac{3p}{\rho}$ $\frac{3 \cdot \text{pressure} \cdot \text{Volume}}{\text{density} \cdot V} = \frac{3p}{\rho}$

• Have N -molecules, each one of them contributing to the pressure

$pV = Nm \langle v^2 \rangle = \frac{1}{3} Nm \langle v^2 \rangle$

• Relating to $\frac{1}{2} (2v)$ in notes

$\frac{1}{3} Nm \langle v^2 \rangle = pV = N k_B T$ Ideal Gas Law

⇒ $m \langle v^2 \rangle = 3 k_B T$

⇒ $\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T$

mean kinetic energy

• So temperature is related to average kinetic energy of the molecules in the gas.

Ideal monatomic gas } $f=3$

$U = \frac{Nf}{2} k_B T = \frac{3N}{2} k_B T$

$\frac{Nk_B T}{P} = V$

$C_v = \frac{3Nk_B}{2}$, $C_p = \frac{3Nk_B}{2} + P \frac{Nk_B}{P}$
 $= \frac{5Nk_B}{2}$

$C_p = C_v + Nk_B$

OR for a mole

$C_p - C_v = R$

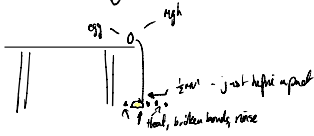
Adiabatic Index

$\gamma = \frac{C_p}{C_v} = \frac{5}{3}$ } for monatomic ideal gas

* Isothermal & Adiabatic processes.

Reversibility: Laws of physics reversible

TEWET?



— Humphry Davy was about reversibility

Heat in a resistor $\sim IR^2 \rightarrow$

* More ways for energy to be distributed as heat than any other way. $\rightarrow$ Statistics of large #s drive many physical processes in an irreversible direction.

Is it possible to carry out a process in a reversible fashion?

* Early researchers realised that if you expanded/compressed a gas quickly it was possible to convert energy into heat $\rightarrow$ irreversibility $\rightarrow$ Shock waves

* However they also realised that if you expanded/compressed slowly so the gas remains in equilibrium throughout the whole process $\rightarrow$ going from 1 equilibrium state to the next seamlessly.

* This is known as a quasi-static process.

Important: In a quasistatic process heat can be emitted/absorbed while maintaining reversibility

*

Reversible process can be run in reverse.

Isothermal expansion of ideal gas

$\hookrightarrow$ constant T $\Rightarrow \Delta T = 0$

$$\begin{aligned} du &= \left(\frac{\partial u}{\partial T} \right)_V dT + \left(\frac{\partial u}{\partial V} \right)_T dV \\ &= C_V dT + \cancel{\left(\frac{\partial u}{\partial V} \right)_T dV} \end{aligned}$$

$$du = C_V dT$$

Recall that $u = \frac{3N k_B T}{2}$

$$\text{so } \left(\frac{du}{dV} \right)_T = 0$$

So for $\Delta T = 0$ we have $\Delta u = 0$.

$$\begin{aligned} du &= dQ + dW = 0 \\ \Rightarrow dW &= -dQ. \end{aligned}$$

"Work done on surroundings equal to heat absorbed"
" " " on gas " " " emitted"

* We now use the expression $dW = -pdV$ and write

$$\Delta Q = \int_{V_1}^{V_2} dQ = - \int_{V_1}^{V_2} dW = \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} \frac{N k_B T}{V} dV$$

$$\Delta Q = N k_B T \ln \frac{V_2}{V_1}$$

"So for expansion $\frac{V_2}{V_1} > 1 \Rightarrow \Delta Q > 0$ "

* What about energy density & pressure.

Same energy, larger volume $\langle \frac{E}{V} \rangle < 0 \Rightarrow p < 0$

Adiabatic Expansion of an ideal gas.

Adiabatic $\rightarrow$ "without flow of heat"

A system with adiabatic walls is said to be thermally isolated.

Any work done produces adiabatic change.

We define **adiabatic** process as one that is both **adiabatic** & **reversible**
so, because adiabatic

$$dQ = 0$$

$\therefore$ the first law then implies

$$dU = dW$$

For ideal gas, $dU = C_v dT$ & using $dW = -p dV$ for a reversible
we have that

$$C_v dT = -p dV = - \underbrace{\frac{RT}{V}}_{\text{for 1 mole}} dV$$

so that

$$\begin{aligned} \int_{T_1}^{T_2} \frac{1}{T} dT &= \int -\frac{R}{C_v} \frac{1}{V} dV \\ &= \ln T_2/T_1 = -\frac{R}{C_v} \ln \frac{V_2}{V_1} \end{aligned}$$

We now have that $C_p = C_v + R$ & divide by C_v

$$\gamma = \frac{C_p}{C_v} = 1 + \frac{R}{C_v} \Rightarrow -\left(\frac{R}{C_v}\right) = 1 - \gamma$$

$$\frac{T_2}{T_1} = e^{(1-\gamma) \ln \frac{V_2}{V_1}} = e^{\ln \left(\frac{V_2}{V_1}\right)^{(1-\gamma)}} = \left(\frac{V_2}{V_1}\right)^{(1-\gamma)}$$

$$\Rightarrow T_2 V_2^{(\gamma-1)} = T_1 V_1^{(\gamma-1)} \Rightarrow TV^{\gamma-1} = \text{constant}$$

Since $pV \propto T \Rightarrow pV^\gamma = \text{constant}$

$V \propto T/p \Rightarrow \frac{T^{\gamma-1}}{p^{\gamma-1}} \cdot T = T^\gamma p^{(1-\gamma)} = \text{constant}.$

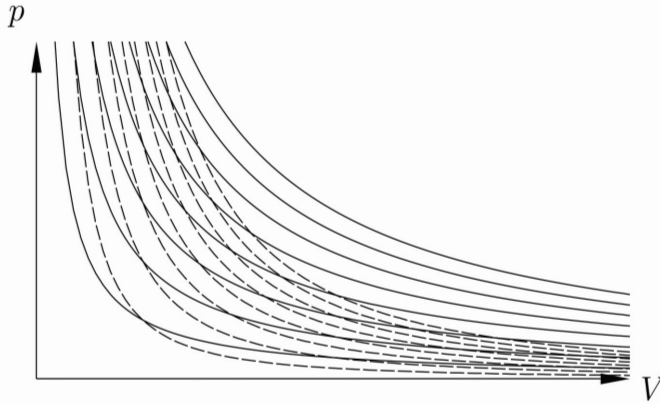


Fig. 12.1 Isotherms (solid lines) and adiabats (dashed lines).